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DEPARTMENT OF PHYSICS

Introduction to Lab Stats

For use in PHYS 151-152 Labs

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August 30, 2025

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Lesson 1

Introduction to Uncertainty

1.1 Opening Remarks

I am confident that everyone taking this course has done a scientific word problem before. More than likely you have done many such problems in your life. With these questions you are given values for several different physical parameters and are asked to find the value of some unknown quantity. In doing all of these questions you have become very skilled in finding *the* answer. There is no debate allowed, no reason to doubt the number that comes up on your calculator. You followed the proper steps, found the solution to the question asked, and everyone in the world should be able to come up with the exact same answer.

However, this kind of infinite precision in your answers becomes impossible to have when we start doing physics in the real world. It is the goal of this manual to introduce you to the ways in which modern science deals with this issue. We will discuss ways of estimating uncertainty in measurements, translating those into the uncertainty in results, and how to properly present results with that uncertainty. In your assignments you will practice the basics over and over again so that you become confident in their usage. In addition, you will be exposed to some more advanced techniques. Mastery of these sophisticated topics is not expected but becoming familiar with them now will enhance your experimental experiences in the future.

1.2 Uncertainty Measurement

In textbook problems you are able to have complete certainty in your answer because the values you start with are exact as well. If a mass of 5 kg is accelerating at a rate of 3 m/s^2 , then the net force on the mass must be exactly 15 Newtons. If we try and conduct an experiment to test this, we will have to measure the mass and the acceleration and those measurements will never be perfect. We will always have some sort of uncertainty in our measurements, and therefore we will have uncertainty in our results.

One rule that many people learn is that the uncertainty in the measurement is half of the smallest scale marking on the measuring device. For example, the uncertainty of a meter stick that is marked every millimeter would be 0.5 mm. However, we cannot rely on such a simple rule. We have to take more things into account about how we measure. Would we

expect a person with good vision to have the same uncertainty in their ruler measurement as someone who could not see as well? Would we expect that the uncertainty in measuring the distance between two thin pencil lines is the same as measuring the distance between two lines made with a broad marker?

Many times, the best practice is to make the measurement several times and create an error profile for that measurement. Later in this manual we will look at the process of transferring the uncertainty in what is directly measured to the uncertainty for a result. This can be a tedious process. Most of the time it is best to simply conduct the entire experimental procedure a number of times and then run statistics on your set of answers. It is with this in mind that the statistical procedures are developed from here on.

Because we will be taking this approach, we will be very liberal in our use of the word measurement; it will not always refer to the direct measurement of a variable. If an experiment is said to measure a value of g , then it is likely that other variables are directly measured and g is the final output of the experiment. However, it is common usage to refer to the experiment as measuring g .

1.3 Significant Figures & Finite Precision

1.3.1 What is a Significant Figure?

A few paragraphs above, we mentioned that a common rule about measuring with a meter stick is that the uncertainty is half of the smallest division marked on the stick, or 0.5 mm. This stems from the fact that when using the stick, the thing you are measuring can be seen to be between two specific millimeter tick marks, and you are then guessing down to the nearest 10th of a millimeter. For example, if you measured the distance between two lines to be 23.46 cm, then you are sure about the 2, the 3, and the 4. However, the 6 would have been a guess because the line you were measuring was somewhere between the 23.4 cm and 23.5 cm tick marks. This gives us the most general definition of what a significant figure is. **When we write down a number, we write down the significant figures of that number, which include all of the digits that we are 100% certain about, plus one digit that is a guess.**

Keep this general rule in mind. No matter if we are talking about recording measurements or calculated values, the goal is to always write down all the digits we are sure of and one digit we are unsure of. Much of what you will be learning in this manual is how to determine which is the first decimal place you are unsure about. In other words the first decimal place of uncertainty.

1.3.2 Tracking Sig Figs Method

One of the simplest ways to determine how many sig figs you should have in your final answer is determine how many sig figs each of your inputs had and then look at the mathematical steps that you took. If the only calculations performed are the addition or subtraction of numbers, the last significant figure in the result occurs at the last decimal place that all numbers in the sum had a significant digit.

$$\begin{array}{r}
 8.16 \text{ m.} \\
 + 74 \text{ m.} \\
 \hline
 82 \text{ m.}
 \end{array}$$

When multiplying or dividing numbers, the number of significant digits in the result equals the smallest number of significant figures in any of the original numbers.

$$\begin{array}{r}
 8.3 \text{ m.} \\
 \times 1,045 \text{ N.} \\
 \hline
 8700 \text{ Nm.}
 \end{array}$$

There are almost no calculations which involve only addition or subtraction, so the rule for multiplying or dividing will be used in all but very rare cases. Perform your calculations carrying all of the digits, then round the final result to the correct number of significant figures.

This last step may be trickier than it seems. In an experimental setting you are constantly making calculations and it is possible that you will need to report any of them. This is part of the reason that computers have become such a large part of modern science. Spreadsheet programs such as Excel allow us to store all of our work and to make sure all calculations are done from the original data. When determining the number of sig. figs. to use you cannot look at the numbers that went directly into the calculation. You must look at all of the raw data that would have an effect on that calculation, even if the raw data was used many steps ago.

1.3.3 Counting Significant Figures

The rules above require knowing how many sig figs any given number has in it. Below are rules for determining how many sig figs a number has, followed by several examples.

- All non-zero digits are significant.
- Leading zeros are never significant.
- Embedded zeros (for example, 506 kg) are always significant.
- Trailing zeros (those to the right of non-zero digits are significant only if the decimal point is shown.
- Counting numbers (for example, 11 apples or 5 bricks) are known exactly; they have an infinite number of significant figures.

Here are some examples,

Number	Number of sig figs
0.003 34 m	3
1.0058 kg	5
500 grams	1
500. grams	3
34.00 Volts	4
0.678 Newtons	3
1.38×10^4 seconds	3
1.3800×10^4 seconds	5
1 meter	1
1 ruler	infinite

1.4 Simple Error Calculations

For basic calculations where no statistical analysis is performed, we can use some simple formulas to evaluate results: Percent Error, Percent Difference, and Percent Change.

Percent Error is used when comparing an experimental result to an agreed upon standard or a theoretical value.

$$\% \text{ error} = \left| \frac{\text{experimental value} - \text{standard value}}{\text{standard value}} \right| \times 100$$

Percent Difference is used to compare two experimental results of the same quantity that were obtained through different methods.

$$\% \text{ difference} = \left| \frac{\text{difference of experimental values}}{\text{average of experimental values}} \right| \times 100$$

Percent Change allows the experimenter to compare before and after results of a variable. Please note that the absolute value is not used here so that the calculation can infer if the change was positive or negative.

$$\% \text{ change} = \frac{\text{final value} - \text{initial value}}{\text{initial value}} \times 100$$

1.5 Formatting Uncertainty Results

When moving beyond the simple tracking-the-amount-of-sig-figs method, the best practice is to quote a plus-or-minus value alongside your result. For example, claiming an event lasted 2.53 ± 0.03 seconds means that you're reasonably confident the event took between 2.50 seconds and 2.56 seconds. In this example the 2.53 would be referred to as the result and the 0.03 would be referred to as the uncertainty.

In general, a plus-or-minus value should be quoted to one significant digit. Think about the rule we stated above that said significant figures are all the ones we know, plus one digit of a guess. The uncertainty values that you will calculate are your estimation of the guess! So we should only get that one significant digit indicating that it is a guess. Think about the absurdity of claiming an uncertainty of ± 0.036 seconds. Who cares about

uncertainty in the thousandths place when we're not even sure what's going on in the hundredths place? That uncertainty is properly reported as ± 0.04 seconds.

Once the uncertainty has been properly rounded, then the actual result should be rounded off to the same position as the plus-or minus value, since that's where the error starts to creep in. Again, this matches up with the rule we stated earlier. We are calculating the uncertainty so that we can find the first decimal place where we would be guessing. So we record both the uncertainty and the result to the same decimal place. The last sig fig in the result is at the decimal place of the first, and only, sig fig in the uncertainty. Thus:

$$\begin{aligned}2.835 \pm 0.08 \text{ s is wrong} \\ 2.835 \pm 0.082 \text{ s is wrong} \\ 2.84 \pm 0.08 \text{ s is correct}\end{aligned}$$

Note that the uncertainty must be fully calculated and formatted before you can determine how to record the result. This is a little unusual as you must determine how to format what is on the right of the plus-minus sign before you determine how to record the what is on the left even though in practice you will usually calculate the result first and the uncertainty second.

One other thing to note is the treatment of order of magnitude and units. The cleanest way, in my opinion, to present this is to treat any power of ten as part of the unit and place both at the end of the uncertainty statement as in the example below.

$$1.78 \pm 0.07 \times 10^{13} \text{ kg}$$

1.6 Types of Error

Before we discuss how we are going to quantify uncertainty, we must make sure we understand exactly what we are attempting to quantify. Errors in experimentation fall into two broad categories: systematic error and random error. Systematic errors are those that tend to bias the results in one particular direction. The main causes tend to be a lack of proper planning and design as well as failure to calibrate devices at the beginning of an experiment. One example might be if a scale was not calibrated to read zero when no weight was on it. Then all mass readings would be either all high or all low depending on which way the scale was unbalanced. On the other hand, random errors arise from things that are much harder to control for. A common source of random error is human error. We can make simple mistakes such as misreading a measuring device or disturbing the experiment, but the key thing to remember here is that those types of errors will be random in nature. It is just as likely that we will make a mistake that gives a higher value rather than a lower value.

All of the calculation techniques that we look at this semester will be an attempt to quantify random error. The lack of focus on systematic error is not because it is unimportant, but rather that these types of error are best dealt with before (and during) data collection. If we take care in our experimental design and make sure to calibrate our instruments, we can eliminate systematic error before we take our data. Once we have our results, we can assume that any uncertainty remaining is due to random error. This will be the approach we will take in this course, but make sure you understand this is an assumption.

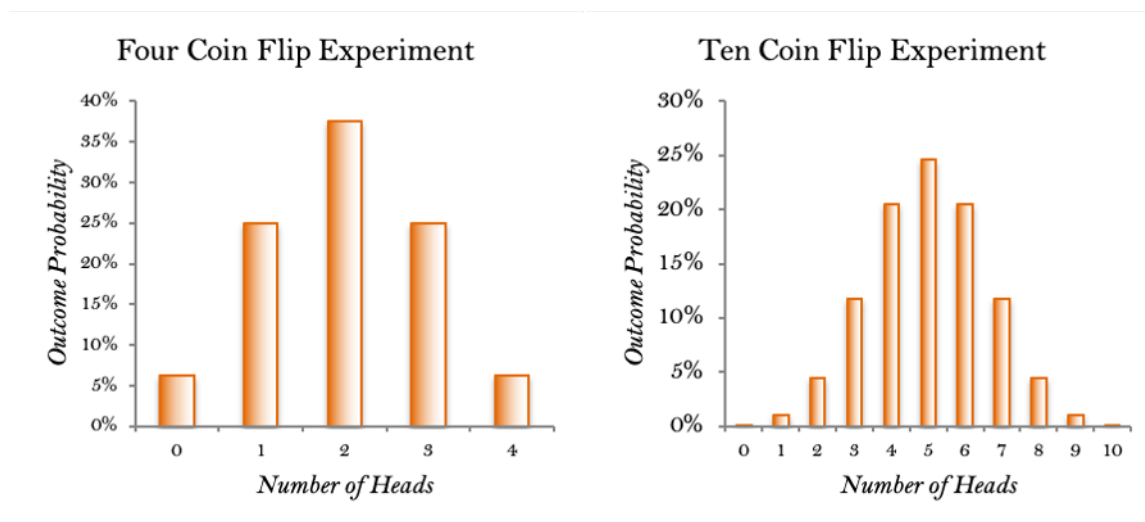


Figure 1.1

There are many techniques that are used to analyze the error analysis itself in order to determine if systematic effects were not taken into account. We leave it to others to pass this knowledge along to you.

1.7 Random Error Distribution

1.7.1 Definitions

In statistics, a quantity that is driven by a random process is known as a random variable. A simple example would be an experiment where you flip four coins simultaneously and count how many coins are showing ‘heads.’ In this case the random variable would be the number of coins showing heads and the random process would be the idea that for each coin there is a 50/50 chance of showing heads or tails. There are five possible outcomes to each trial: no coins will show heads, only 1 will, 2 will, 3 will, or all four will. If this experiment is performed a large number of times the results of the experiment can be summarized on the left side of Fig. 1.1.

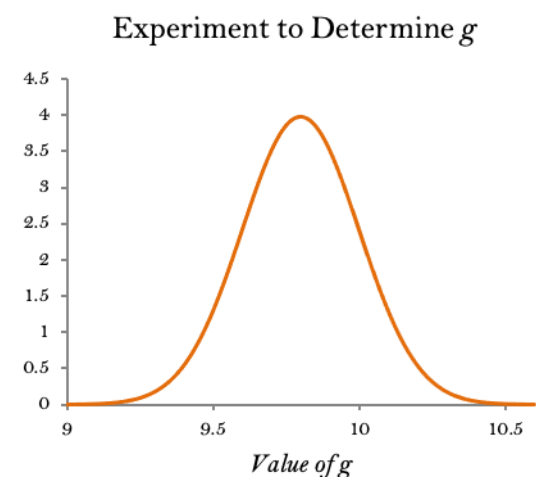


Figure 1.2

This distribution should not surprise us. There is a relatively small chance that we would get either all heads or all tails, and there is a high probability that we get exactly half the coins to show heads. If we change the experiment to throw 10 coins at once instead of 4 then we get a similar pattern shown in the right panel of Fig. 1.1. The chances of getting all heads or all tails is virtually non-existent, and the highest probability is still that half of the coins show heads.

In this experiment the number of coins showing heads is specifically called a dis-

crete random variable. That is, only specific values for the variable are possible. There is no way, for example, to have 3.5 coins show heads. A more realistic physics example might be an experiment that attempts to measure the value of g . This would be known as a continuous random variable as the experiment could produce any value for g . In this case the underlying random process would be the fact that there is random error present in all of the measurements that are taken to produce the final result. The distribution for a continuous random variable will simply be an extension of the two charts already shown. Imagine what the graph would look like if 100,000 coins had been tossed at once.

The curve in Fig. 1.2 gives us the probability distribution of a continuous random variable. It is sometimes referred to as the bell curve given its shape. However, several different distribution curves have a bell shape to them, so this term is ambiguous. The proper term for the function is the Gaussian Distribution. It continues to have the same important properties as the plots for the discrete random variables: there is a high probability for getting a result close to some central value and a low probability of getting a result very far beneath or above the central value. You may notice that the y-axis is no longer in percentages and that values for the curve go above 1. This means that we cannot simply find the probability of a particular outcome by knowing the value of the function at some given x-value. We will discuss this and the other mathematical values that define this function in the following sections.

1.7.2 Central Value

The most obvious feature of the Gaussian Distribution is the peak of the curve around the central value. This central value is of course the average of all possible values of the variable in question. It should be of no surprise that the average plays a large role in defining a curve that is the consequence of random error. The whole reason that we take an average is to eliminate random error. We know that if we only measure something one time the chances of us getting the exact right value are very low. We feel much more confident in taking the measurement several times and reporting the average value. This increase in confidence is because we have eliminated some of the random error from our result.

As you may know there are several different ways to calculate the average value of a data set. The most common are the median, the mode, and the mean¹. Consider the following data set.

$$\{1, 2, 2, 3, 4, 7, 9\}$$

The median is defined as the middle number of the set when they are arranged in numerical order as above. The median of this set is 3. If a set has an even number of data in it then the median will be the value that is in-between the two middle numbers.

The mode of a set is the number that appears most often in a set. In the example above the mode is 2.

The mean is the most common way to calculate the average. It is the sum of all numbers in the set divided by the number of values in the set. For our sample set, the mean is 4.

¹There are actually several different types of means as well, the arithmetic mean and the geometric mean being the most common. Throughout this text when we say mean we are referring to the arithmetic mean.

The formal way of writing the mean is,

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} \quad (1.1)$$

Please note that we will use the notation of an x with a bar over it for an average value. This is sometimes pronounced ‘x bar’. It is also important to note that the mean, median, and mode of a Gaussian curve will be the same value. This is not true for all probability curves.

1.7.3 Spread in the Data

Of course, the average is not the only number that can describe a data set. It would be possible to have many different data sets with the same average but that are very different from each other. Take the following two sets as examples,

$$\begin{aligned} \text{Set } A &= \{98, 100, 102\} \\ \text{Set } B &= \{0, 100, 200\} \end{aligned}$$

Both of these sets have an average value of 100, but if these were results from an experiment, we would certainly feel that the data from Set A was much more reliable than the data from Set B due to the differences in spread of the two sets. The most common way of calculating such a spread in the data is the *standard deviation*, which is defined as²,

$$S = \sqrt{\frac{\sum_{i=1}^N (\bar{x} - x_i)^2}{N}} \quad (1.2)$$

For our two set we would have,

$$\begin{aligned} S_{\text{Set A}} &= \sqrt{\frac{(-2)^2 + (0)^2 + (2)^2}{3}} = \sqrt{\frac{8}{3}} = 1.633 \\ S_{\text{Set B}} &= \sqrt{\frac{(-100)^2 + (0)^2 + (100)^2}{3}} = \sqrt{\frac{20,000}{3}} = 81.65 \end{aligned}$$

The idea is to take the difference between each data point and the average of the set, square all of the differences, add the squared values together, divide by N , and then take the square root. The differences are squared so that negative deviations will not cancel out positive ones³. Taking the square root will ensure that the standard deviation will

²Some of you may have seen a formula where the denominator is $N - 1$. We will discuss this difference later on.

³You may ask why not just take the absolute value? It turns out that if you start with the assumption that each measurement is truly random, you can derive a function that gives the Gaussian Distribution curve. One of the defining parameters of this function is the standard deviation as we have defined it. There is a statistic called the *mean absolute deviation* which looks very similar to the standard deviation,

have the same units as the original data, which makes them easier to compare. If you do not take the square root, then you have a quantity called the variance.

$$\text{Variance} = S^2$$

Our two examples are small data sets, but there is an important feature to the standard deviation that occurs with larger data sets. If we have an experiment that we perform many times, the standard deviation will be the same no matter how many times the experiment is repeated. That is the formula above is not dependent on N . At first this may seem counter intuitive, but let's take a closer look. As more trials are performed N will go up, so the denominator will get larger. However, each time another trial is conducted the sum in the numerator will get an additional term. Once more, this additional term will be positive, (or 0), as it is always squared. Most of the time the numerator will grow only a little as most trials will be near the average. However, occasionally, a large deviation will occur that will offset all of the small deviations. In this way both the numerator and denominator grow linearly with N , and so N will have no overall effect.

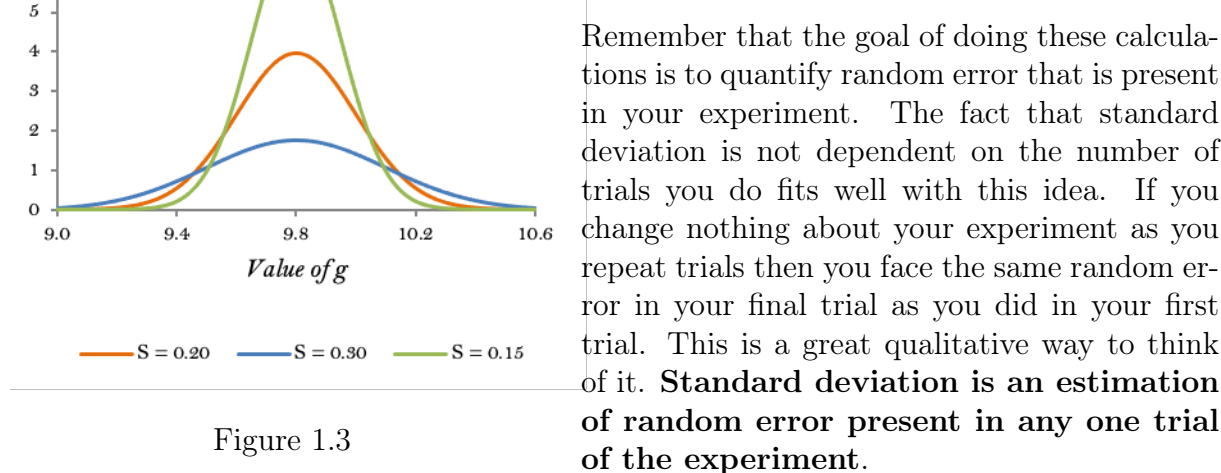


Figure 1.3

Remember that the goal of doing these calculations is to quantify random error that is present in your experiment. The fact that standard deviation is not dependent on the number of trials you do fits well with this idea. If you change nothing about your experiment as you repeat trials then you face the same random error in your final trial as you did in your first trial. This is a great qualitative way to think of it. **Standard deviation is an estimation of random error present in any one trial of the experiment.**

1.8 Standard Deviation & the Gaussian Distribution

The standard deviation will also help define the Gaussian Distribution and the relationship between the two will help us make sense of our uncertainty results. Figure 1.3 below shows three different Gaussian curves all with a peak value of 9.8, but with different standard deviations.

The change in standard deviation causes a change in the width of the curve. The smaller the deviation, the more narrow the peak. In order to interpret these different curves, it is important to understand how we get probabilities off of the curves. Earlier we said that the y -axis values were not percentages. Part of the reason for this is that each of the curves in Figs 1.2 and 1.3 are normalized so that the area underneath the curve is exactly equal to 1. The way to interpret this is to say that the probability that a single

except that it takes the absolute deviation instead of worrying about squaring each one and then taking the square root at the end. While this also gives an estimation of the random error present in each trial, the probability distribution would be slightly different than the Gaussian curve, and therefore, it would not be based on the assumption of pure randomness in each trial.

% of trials lie within...	...standard deviations of the average value
68.27	1.00
95.00	1.96
99.00	2.58
99.73	3.00
99.994	4.00
99.99994	5.00

Table 1.1

trial of the experiment will fall between ∞ and $-\infty$ is 100%. This is a true, but not very illuminating statement. However, similar statements can be made with different boundaries. If I wanted to know the probability of finding a result between 9.4 and 10.2 in one of the curves in Fig 1.3, then I would need to find the area under the curve between those two points. This can of course be done with integration, but luckily the results are quite standard and are summarized in Table 1.1.

For example, if you are performing the experiment characterized by the orange line in Fig 1.3, then 95% of the trials you do will lie within ± 1.96 standard deviations of the average. That is 95% of your trials will lie within ± 0.392 , ($1.96 * 0.20 = 0.392$), of 9.8 or 95% of your trials will lie between 9.408 and 10.192.

This same statement can be reversed. If another trial is performed then you can be 95% sure that it will fall within that same range. This is an interesting statement, but it doesn't quite get us to what we want. If you performed this experiment, you would want to report your result as $g = 9.8$, but you need to also publish what the uncertainty in that average value is. The standard deviation cannot do this. It can only tell you about the random error associated with a single trial of the experiment.

1.9 Standard Error

1.9.1 Definition

Let's review a few thoughts before we continue. We know that it is better to perform multiple trials and report the average of those trials rather than only doing it once. We know that this is because random error is reduced in the process of averaging. This should mean that any estimation of uncertainty in the average should be less than the uncertainty in any given trial. We would also feel more confident about an average that came from a large data set compared to a small one. The calculation of **standard error** will fit these ideas quite well.

$$\bar{S} = \frac{S}{\sqrt{N}} \quad (1.3)$$

The standard error is simply the standard deviation divided by the square root of the number of trials performed. This formula will ensure that the standard error is always less than the standard deviation and that as the number of trials goes up, the standard error will be reduced. Therefore, the correct way to calculate uncertainty for an average

is to use the standard error. For this reason, the standard error is also referred to as the *standard deviation of the mean*.

Now these properties could be obtained if the denominator was N or N^2 or any other variety of factors dependent on N . Of course, it can be mathematically shown that the root of N is correct, but for intuition we can look to the law of diminishing returns. If you wanted to cut your level of uncertainty in your average value by half, you would have to do 4 times the number of trials. If you want to be certain of your average to an additional decimal place you have to do 100 times the number of trials! So there is great advantage in performing more than a small number of trials, but eventually the marginal improvement on your uncertainty become relatively small.

1.9.2 Standard Error & the Gaussian Distribution

Let's think about how to get the uncertainty of the average in a different way and return to our experiment to measure the value of g . We could perform 10 trials of the experiment and get an average of those 10 trials. Maybe we get an average of 9.82 m/s^2 . We could then do another ten trials and perhaps for the second set we get an average of 9.79 m/s^2 . The point is that we would not expect the average of any set of 10 trials to be exactly the same. There would always be some small variation. However, we would expect that variation to be smaller if in each set the number of trials was increased, say to 50. So again, we can see the influence of the number of trials on the uncertainty of the average.

What is most important about this method of sampling is that the different averages, (for a constant number of trials), will be randomly distributed. So if we plot the frequency at which we see each particular average value, we produce Gaussian Distributions as shown below.

The orange line is the same distribution as it was in Figs 1.3 and 1.4. We have simply reduced the domain of the x-axis. Keeping in mind that the area under the curve represents the probability of finding a value in that range, notice how few individual trials would fall between 9.6 and 10.0. However, we see that if we take the average of ten trials, the probability of finding an average in that range, (area under the blue curve), increases dramatically. Of course, the area under the green curve is higher still and so the probability of getting an average of 20 trials in this range is larger as well.

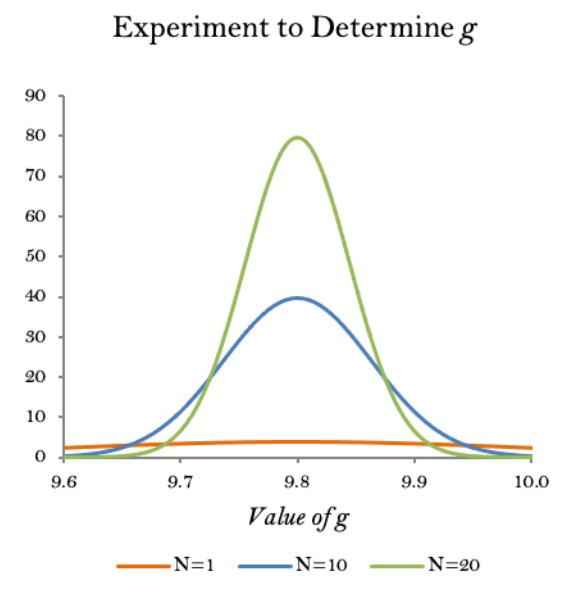


Figure 1.4

As these curves are Gaussian, the mathematical properties are the same as the ones we discussed earlier. Here the central value would be the average of the averages taken, which we will assume is the true value of g and the width of the curve is controlled by the standard error. We can also feel free to use the numbers listed in Table 1.1 above. For the green curve $S = 0.20$ and $N = 20$, so $\bar{S} \approx .045$. This means that we could say

that our experimental value for g is 9.8 and we can be 95% certain that the true value of g is within ± 0.088 ($1.96 \times 0.045 \approx 0.088$) of 9.8. Written in proper format our result would be $g = 9.80 \pm 0.09 \text{ m/s}^2$.

At last we have a way of to present uncertainty for our experiments. We would perform multiple trials and present the average value of those trials as the result with the standard error as basis of uncertainty present in the average value. Note that you need to make clear what confidence level you are presenting your results to. Talk to your lab instructor and see what they prefer. When in doubt specify the confidence percentage along with your result. This is something that we will use when reporting results throughout the semester so please keep this in mind as you perform your lab work.

1.10 One Final Adjustment: Population & Sampling

Let's look at a simple example. Say I wanted to know the average height of the students in Monday's lab class. I could certainly measure everyone and calculate an average and standard deviation. Assume I get the following results (ignoring formatting rules for the moment).

$$\begin{aligned} \textit{Average} &= 175.2 \text{ cm} \\ S &= 3.47 \text{ cm} \\ N &= 24 \end{aligned}$$

In this case the standard deviation gives me an idea of what the spread about the average is. The standard error could be calculated, but it would not have much meaning.

Things get complicated when I want to assume the average height of people in this class represents the average height of all students at Emory. If this assumption is to be in any way valid, I would need to put a great deal of effort into making sure that the group of students I use is representative of the Emory student population. This is no easy task. There is a whole field of study devoted to experimental design, and while it will not be something we spend a great deal of time on this semester, if you become a professional scientist, you will certainly have to consider design issues carefully when creating your own experiments. For now, just understand that error can sneak into your results simply by observing only a sample of all possible data.

Now in a physics setting we are attempting to measure physical parameters and therefore we can never measure the full population of values. People could measure values of the speed of light for a billion years and yet they could still keep measuring; the population size is infinite. Therefore, it will always be appropriate to consider the measurements we do as a sample of that infinite population.

It can be shown that if we assume that what we are measuring is a random variable, and that each trial is independent of each other one, that increase in error due to taking a

sample is dealt with by modifying the standard deviation formula.

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (\bar{x} - x_i)^2}{N - 1}} \quad (1.4)$$

We will use the Greek letter sigma to represent the unbiased standard deviation of a sample. We say unbiased because if the denominator was simply N as it was before, our estimation of the standard deviation would be too low. By decreasing the denominator, the estimation of error goes up and this corresponds to the increase in random error introduced into our experiment by sampling.

We can now put some meaning to the numerical results we have. For the simple experiment listed above, we could say...

- $S = 3.47$ cm and this represents the spread of the heights around the class average.
- $\sigma = 3.54$ cm and I could be 68% sure that any one random Emory student falls this far above or below the class average of 175.2 cm.
- $\bar{\sigma} = 0.724$ cm and I could be 68% sure that actual average height of Emory students is this far from the measured sample average of 175.2 cm.

So, in the end results can be summarized as such. There is some true value that we are trying to measure. Our best guess at that true value is the average of our sample data set and the unbiased standard deviation will give us our uncertainty in our average value once Table 1.1 (or a similar table) is used to convert the standard deviation into uncertainty. Also note that once the standard deviation is corrected for sampling bias, nothing changes about the standard error formula. The unbiased standard error is simply,

$$\bar{\sigma} = \frac{\sigma}{\sqrt{N}} \quad (1.5)$$

One last word about calculating these values in Excel. There are three standard deviation formulas in the program and here is how they should be used.

`=stdev()` or `=stdev.s()`: Both of these functions give the unbiased standard deviation of a sample. This is what we will use the majority of the time.

`=stdev.p()`: This gives the standard deviation of the population. This is the formula from several pages ago when we first wrote an expression for standard deviation, and it should only be used if your measurements encompass the entire population, not just a sample of it.

Lesson 2

A Few More Basics

2.1 Basic Error Propagation

So far we have looked at getting the uncertainties and errors in individual quantities. Of course many times we have to combine separate results together in order to get our final result. There are special rules for combining the errors of different quantities and this process is known as error propagation. We will spend an entire lesson on this topic soon, but for now let's focus on the most basic rule. Assume I have a very basic function where I am only multiplying a physical parameter by a constant,

$$A = cB$$

If we know the error in B , then we simply multiply the error by c to get the error in A .

$$\sigma_A = c \times \sigma_B$$

(Here σ stands in for any of the uncertainty calculations talked about in Lesson 1: standard deviation or standard error). For instance, imagine I took the data from lab M1 and plotted the ball's y-position vs. time-squared. Given the formula $y = 1/2gt^2$ the slope of such a plot should equal $1/2g$. Let's say I get a value for the slope of 4.9 ± 0.0463 (ignoring formatting rules for the moment). You can get an estimation of g if you multiply the slope by 2. This means that the plot gives an error value for g of about 0.0926. Formally I could say that the graph gives me the result of $g = 9.8 \pm 0.2 \text{ m/s}^2$ with a 95% confidence interval.

Many times equations are plotted in ways that may at first seem unusual. By plotting variables so that we have a linear relationship, and that the slope differs from our desired result only by a constant factor, we can easily obtain results with uncertainties. Let's now discuss how we can find the error in the slope.

2.2 Quick Error Calculation for a Linear Plot

In the coming weeks we will look at advanced Excel techniques that will allow you to get equations for a wide variety of functions as well as the errors that go along with the parameters of those equations. However, many times we simply need the slope value and

it can be handy to know how to calculate the uncertainty of the slope straight from the graph. This can be achieved with the following formula.

$$\sigma_{slope} = slope \times \sqrt{\frac{\frac{1}{R^2} - 1}{N - 2}} \quad (2.1)$$

So this formula can give the standard error of the slope once you have the slope itself, the R^2 value, and the number of data points in the set. Most of the time in this class it will be fine to just use the rounded values of the slope and R^2 that are displayed on the graph. You can get more exact values by using the following two formulas. These can also be handy if you just want the information without actually plotting a graph.

$$=slope(known\ y's, known\ x's) \qquad =rsq(known\ y's, known\ x's)$$

2.3 Degrees of Freedom

As we saw at the end of the first lesson, there is an increase in our error if we are only measuring a sample of the full population of data (remember this is always the case in Physics). This effect was made manifest by changing the denominator of the standard deviation function to $N - 1$ from N . In general we say that the denominator of both equations is the number of degrees of freedom in the calculation. If you measure the full population, then the degrees of freedom equals the number of data points. If you take a sample, then the degrees of freedom is reduced by one for calculating the standard deviation. In general the degrees of freedom is calculated as the number of data points minus the number of parameters that must be estimated in doing the calculation. As an example, when we calculate the standard deviation, we must first estimate the true value, which we do so by calculating the average of our data set. This is the only estimation in the calculation; the only other numbers in the calculation are the data points we have measured. So one estimation reduces the degrees of freedom by 1.

Now think about the equation you get when you plot a linear trendline. That equation contains an estimated slope and y-intercept value, and together they represent the model that you are fitting to the data. So because that model contains two estimates, (slope and y-intercept), the degrees of freedom is $N - 2$. The number of degrees of freedom not only effects the equations of error themselves, but it also effects how we adjust for different confidence intervals.

2.4 Small Number Statistics

So far we have been using the values in Table 1.1 of the first lesson to convert our errors into uncertainties at specific confidence intervals. This assumes that our data will fall under the normal distribution. Unfortunately, this assumption is not a very good one when we are only taking small amounts of data. With the experiments that you perform in lab, you end up taking a fairly small number of data points simply because we only have three hours in which to conduct the experiment. Luckily this issue is easily solved by having a different table of multipliers.

Table 2.1 lists the numbers needed to convert errors into uncertainties for a variety of confidence intervals. Note that it is dependent on the number of degrees of freedom in the estimation, not the number of data points. The numbers in this table are calculated from the Student's T-Distribution. As the number of df increases, the Student's T-Distribution will approach the Normal Distribution, and at $N = \infty$ the two are equivalent. In practice, it is common to use the normal distribution once the df reaches 100. You will notice that the multipliers on the last row of Table 2.1 match the numbers on Table 1.1.

df	68.27%	90%	95%	99%	df	68.27%	90%	95%	99%
3	1.20	2.35	3.18	5.84	24	1.02	1.71	2.06	2.80
4	1.14	2.13	2.78	4.60	25	1.02	1.71	2.06	2.79
5	1.11	2.02	2.57	4.03	26	1.02	1.71	2.06	2.78
6	1.09	1.94	2.45	3.71	27	1.02	1.70	2.05	2.77
7	1.08	1.89	2.36	3.50	28	1.02	1.70	2.05	2.76
8	1.07	1.86	2.31	3.36	29	1.02	1.70	2.05	2.76
9	1.06	1.83	2.26	3.25	30	1.02	1.70	2.04	2.75
10	1.05	1.81	2.23	3.17	35	1.01	1.69	2.03	2.72
11	1.05	1.80	2.20	3.11	40	1.01	1.68	2.02	2.70
12	1.04	1.78	2.18	3.05	45	1.01	1.68	2.01	2.69
13	1.04	1.77	2.16	3.01	50	1.01	1.68	2.01	2.68
14	1.04	1.76	2.14	2.98	55	1.01	1.67	2.00	2.67
15	1.03	1.75	2.13	2.95	60	1.01	1.67	2.00	2.66
16	1.03	1.75	2.12	2.92	65	1.01	1.67	2.00	2.65
17	1.03	1.74	2.11	2.90	70	1.01	1.67	1.99	2.65
18	1.03	1.73	2.10	2.88	75	1.01	1.67	1.99	2.64
19	1.03	1.73	2.09	2.86	80	1.01	1.66	1.99	2.64
20	1.03	1.72	2.09	2.85	85	1.01	1.66	1.99	2.63
21	1.02	1.72	2.08	2.83	90	1.01	1.66	1.99	2.63
22	1.02	1.72	2.07	2.82	95	1.01	1.66	1.99	2.63
23	1.02	1.71	2.07	2.81	≥ 100	1.00	1.64	1.96	2.58

Table 2.1: Multipliers for various statistical degrees of freedom and confidence levels. These multipliers are for two-tailed statistical tests. For example, if you select the 90% confidence level, then you are assuming that 5% of the data will fall below the confidence interval calculated, and 5% will fall above that interval.

Lesson 3

Introduction to Regression

3.1 How is a Curve of Best Fit Determined

So far we have looked at how to estimate uncertainty when you have a number of measurements of the same quantity and when you have simple linear graphs. As you can imagine, not every phenomenon in the world can be represented by a linear graph. In this lesson we will look at how computers calculate trendline equations and how to get uncertainties for the parameters of those equations. The methods you will learn will allow you to gain access to these uncertainties for more complicated functions.

Let's start by thinking of a set of non-linear data points plotted on an xy -graph. If we were doing this by hand we could estimate by eye what a best fit curve would be. We could then try and quantify how well this curve fit the data by looking at the vertical distance between the each point and the best fit curve. We might then average these distances, but we would run into the same issue as with standard deviation; we do not want the negative distances to cancel out the positive ones. So again, the solution is to use the squared distances.

This technique is known as the method of least squares and it can be used for any plot no matter how strange the curve gets. The computer will try many different possible curves to fit the data and for each one it will look at the sum of the squares of the differences between the curve's y -values and the data points' y -values. The curve that is chosen as the best fit will be the one where this sum is the smallest.

You may ask why we are looking only at the vertical when we could look just as well look at the horizontal or even at a line that is the shortest distance between the point and the curve. This comes down to the idea that the x -variable is independent. As experimenters we have control at which x -values we take our data at, so it is assumed that any error that exists will be in the dependent variable on the y -axis. Now of course this is just an assumption, but in most cases it is a good one.

The procedure that is described above is a very general one and is known as a numerical technique. The computer guesses at a solution, sees how good it is, and tries to make better guesses until it can no longer improve the solution. For simple curves like lines and parabolas there are analytic solutions, meaning the equation parameters can be directly calculated from the data points; no guessing needed. So when you plot on Excel, there is a formula that gives the equations presented. However, this exact answer is the same

solution you would get if the machine used the numerical method to find it. One advantage to the analytical solution is that every program or computer should give the same solution for the same data set (think Excel vs. a TI calculator). It is the outcome of these analytic methods that we will be looking at in this lesson.

3.2 Excel Methods

In order to demonstrate how to get Excel to calculate the uncertainties in equation parameters we will work with a specific data set. You will want to download the file entitled "Projectile Motion Data" that is posted on the lab's Canvas page. The idea is to follow along with the steps listed below as you go through this lesson. We will start to analyze this data here, but we will return to it several times over the semester to make improvements to our estimates.

When you open the file you will see data from an experiment similar to lab M3. The major difference is that this data was collected in a manner that doesn't involve the issues that arise from a rolling shutter. So it is OK if we have all of the up-trip and down-trip data on the same plot. Like in lab, the x and y positions of a projectile were measured every 30th of a second. Let's start by examining a graph of x -position vs. time.

- Create a scatter plot of x vs. t as you did in lab and add a trendline. You should get $y = 3.0552x - 0.0009$.
- Now click on an empty cell somewhere on the sheet. You are going to use an array formula, which means it will have multiple outputs. This means you need to make sure that your empty cell is at the top left of a blank area that is 2 columns wide and five rows high.
- Now you are going to use the formula `=linest()`.
 - Type `=linest(`
 - Highlight the x -position values and type a comma.
 - Highlight the time values and type a comma.
 - Finish the formula by typing `"1,1")` and hitting ENTER. Your formula should look like this... `=linest(C4:C25,E4:E25,1,1)`
- The right side of Table 3.1 shows the output that you should get¹. The left side of Table 3.1 shows the meaning of each value of the output.
- Of course we see the slope and y -intercept of our line in the top row. Note that we get the full, non-rounded values of these parameters and can click on them to use in other Excel formulas.
- The second row gives us the standard error for both the slope and the intercept. These errors are equal to an uncertainty around 68% confidence. You would multiply these errors by values from Table 2.1 in the previous lesson to get uncertainties at higher confidence.

¹If you only get a single number as an output it may be because you have an older install of Excel. You will want to highlight a blank area of the sheet that is 2 columns wide by 5 rows tall. Re-type the formula and hit CTRL+SHIFT+ENTER.

3.05518916	-0.0008617	slope	y -intercept
0.0103371	0.00422713	slope error	intercept error
0.9997711	0.0102535	R^2	se-regression
87353.1176	20	F	df
9.18379677	0.00210268	ss-regression	ss-residuals

Table 3.1

- The third row tells us how well the regression model fits the data. The R^2 value is the same as on the graph. This is a value that can range from 0 to 1. An R^2 of 1 would mean all of the data points fit on the trendline and that the variation in your x values completely explains the observed variation in the y values.
- The *se-regression* is the standard error of the regression. It is the quantification of the general regression technique that we were discussing earlier. If you were to take the average squared distance between the predicted y -values and the actual y -values and then take the square root, you would get this value. The interpretation is that about 68% of the data points should fall within this amount from the trendline. Again Table 2.1 can be used to get other intervals. Please note that the standard error of the regression has units of the y -values.
- In the final two rows, the only number we will care about is df , which is the number of degrees of freedom. Here we have $N = 22$ and we are estimating two parameters: the slope and y -intercept. So in this case $df = 22 - 2$ or 20.

The *linest* function has the following form, $=\text{linest}(\text{known } y\text{'s}, \text{known } x\text{'s}, \text{intercept}, \text{stats})$. In the example above we wanted to plot the x -position on the y -axis, so we highlighted it first. We then highlighted the times as they were to go on the x -axis. We put in a 1 for both the intercept and stats argument in the formula. These are binary arguments so they can be 0 or 1. The intercept argument is wanting to know if it should include an intercept or not. As we wanted an intercept we entered 1, if we didn't want one we would have put 0. Likewise, because we wanted the additional statistical information we put a 1 for the stats argument.

3.3 Beyond $y = mx + b$

We can use the *linest* formula for some more complicated functions, like polynomials. You should know by now that the y -position is related to time through a second order polynomial. Let's use this as our next example.

- Again, start by making a plot for comparison. A graph of y -position vs time should give you a fit equation of $y = -4.9214x^2 + 3.805x + 0.6793$.
- Calculate a column next to the time column that is time-squared.
- Select a cell that is at the top left hand corner of a blank area that is 3 columns wide by 5 rows tall. We need the extra row because this model has three parameters in it.

- We will use the *linest* formula again. This time highlight the *y*-positions as the *known y's*.
- Highlight both time and time- squared at once as the *known x's*.
- Put a 1 for both the *intercept* and *stats* arguments.
- You should have the formula =LINEST(D4:D25,E4:F25,1,1)
- Hit ENTER to complete the array formula. You should get the output displayed in Table 3.2.

−4.9213509	3.80501341	0.6729545
0.04633656	0.03359179	0.00507487
0.99853819	0.00866664	#N/A
6489.30402	19	#N/A
0.97483072	0.0014271	#N/A

Table 3.2

The table output is similar to the first graph; the first two rows contain the values of three parameters of the polynomial equation and their associated errors. The final three rows of the first two columns are the same statistics as in Table 3.1, while the last column of these rows are blank. Note that the degrees of freedom has dropped to 19 as we are now estimating three parameters.

One confusing thing to note here. As far as the function *linest* was concerned you had two *x*-variables; time and time-squared. If you followed the instructions exactly you would of had your time data in column E and your time-squared values in column F. These columns had to be next to each other in order for Excel to work, but Excel would not care which column was first. The first column of the output, (at least the first two rows), corresponds to the *last* column selected as part of the *known x's* parameter in the *linest* formula. That is −4.921... is the parameter in the curve-fit that you multiply t^2 by and 0.04633... is the uncertainty in that parameter. The second column corresponds to the variable t . The last column contains information on the intercept. So, you've seen *linest* used with both 1 and 2 *x*-variables, but of course you could use more. Just remember that the first column of the output corresponds to the last column highlighted for the *known x's* and that going right through the output, corresponds to going left through the highlighted *x*-variables. Not how I would have programmed it, but there it is.

Your quiz this week will mostly be more practice of this technique, (no worries only 1 or 2 *x*-variables). Can you guess what graph you might be looking at on the quiz?

Lesson 4

Interpretation of Results

4.1 Overlapping Uncertainty Ranges

So far we have looked mainly at how to calculate errors and uncertainties in our results. To be able to use these calculations intelligently, we must make sure we understand exactly what these results mean. In the first lesson we discussed several basic points. First, we said that all of these techniques are looking to quantify random error. This is because we assume that we do our best to eliminate systematic errors and that the random ones are the errors we cannot eliminate. We also stated that the standard deviation of a set was the amount of random error present in a single value in the set. Finally we presented the standard error as the amount of random error left after taking the average of a set. This was later expanded to include other statistical processes, namely regression.

But what if I have two results. Let's say we have measured the spring constant of a spring through two different methods and obtained the results $k_1 = 6.72 \text{ N/m}$ & $k_2 = 6.84 \text{ N/m}$. If I were to ask the simple question, does $k_1 = k_2$, then the answer would be no. Those two numbers are clearly not equal to each other. However things get a bit more complicated if I ask if the following two quantities are the same,

$$k_1 = 6.72 \pm 0.09 \text{ N/m} \quad \& \quad k_2 = 6.84 \pm 0.07 \text{ N/m}.$$

The answer isn't no, but it isn't yes either. Given these uncertainties then the magnitude of k_1 is from $6.63 - 6.81$ and the range of k_2 is from $6.77 - 6.91$. These ranges overlap, so what we can say is that the two values *could* be the same. Another way to put this is that the two experiments do not give contradictory results. What we could conclude is that either method is a valid way of finding the spring constant. If the ranges did not overlap, then we would have to examine the physics of each experiment in detail to see where systematic error could be creeping into the results. However, we would have to examine both methods as we would not know which one was 'correct'.

Let's examine another quick experiment. Say we measure the time it takes for a heavy object to fall from a height of 1.5 meters and we get the result $t_{heavy} = 0.56 \pm 0.04 \text{ s}$. We then time drops from the same height for a much lighter object and get a result of $t_{light} = 0.53 \pm 0.02 \text{ s}$. What conclusions can be reached from this data? We cannot say that the drop times are the same, but we also cannot say that they are different. We have failed to prove that the time it takes to fall this distance is dependent on mass. This indicates that the time is independent of mass, but this single experiment does not prove

that. Now of course experiments like this have been repeated over and over again, and the overwhelming majority of these experiments give results where drop times do not have a statistically significant difference. It is the weight of all of this evidence together that allows us to confidently say that the time is independent of mass. However, we can never technically prove it 100%. There is always a chance that a new experiment will show a mass dependence for the drop time. While that outcome is very unlikely, when reading results that are closer to the frontier of science it is good to keep in mind that a single result, even a significant one, can be contradicted by other statistically significant results. If you report all your results to 95% uncertainty, then on average, 1 out of every 20 results you publish will be a false positive.

There is one special comparison that we need to touch on, and that is the comparison to zero. Let's say we get a result for some quantity that is $x = 4 \pm 5$. There are a couple of things to note here. First of all, we only know x to one sig. fig., so we are not getting a lot of detail out of this experiment to begin with. Secondly, and more importantly, the uncertainty is larger than the result, meaning that it is possible that $x = 0$. If we extend the reasoning that we were talking about earlier, this means we have failed to prove that x is not zero. A more typical way of saying this is that we can assign no value to x with any statistical significance.

Lesson 5

Propagation of Errors

In the previous lessons we have looked at how to calculate the uncertainty in our results when we have a collection of results. That is we looked at calculating the uncertainty of an average value and the uncertainties produced by plotting data on a graph. These are the most important techniques to know, because they are the most widely used. However, in some cases you will not have a large amount of data to work with. In this case we need to know how the errors in your inputs become the errors in your outputs. This technique is known as propagation of error.

5.1 Propagation of a Constant

We have already seen the simplest form of propagation where the outcome, f_1 , is simply the input, x , multiplied by some constant value, A . That result was of course...

$$\begin{aligned} f_1 &= Ax \\ \sigma_{f_1} &= A\sigma_x \end{aligned} \tag{5.1}$$

For this lesson, the Greek letter sigma will stand for any type of error, (standard deviation, standard error, regression error, ect...).

5.2 Propagation Through Addition & Subtraction

We now want to focus on functions that have multiple inputs, each of which have their own uncertainties. The most important thing to keep in mind is that error always goes up. If I have two variables that contain error, there is no way in which I can combine them and the resulting error is less. This is of course assuming that the error in all of my inputs are random and independent of one another.

Let's look at the next simplest case, addition.

$$f_2 = x + y \tag{5.2}$$

Now while it is true that we cannot assume that the errors in x and y will completely cancel each other out (which would be rather fortuitous), the random nature of the errors means that some cancellation happens, and we do not simply add them either. Instead we can treat the errors like vectors. When examining motion in two dimensions you know that motion in the horizontal direction is independent of motion in the vertical direction. This matches nicely with our assumption that our errors in x and y are independent. So we will combine the errors just like we would combine the x and y parts of any vector quantity you know.

$$\begin{aligned}\sigma_{f_2} &= \sqrt{\sigma_x^2 + \sigma_y^2} \\ &\text{or} \\ \sigma_{f_2}^2 &= \sigma_x^2 + \sigma_y^2\end{aligned}\tag{5.3}$$

Remember from the first lesson that the square of the standard deviation is called the variance. Again, for this lesson we will refer to the square of any type of error as the variance. In many cases it turns out to be easier to speak in terms of variances when doing propagation of errors. Calculations also tend to be easier when working with variances, but you just have to remember to take the square root at the very end to make sure you go back to the correct units.

Because we are treating these errors like vectors, then the expressions above are also true when we are doing subtraction. That is the spread that we could see in our result is the same whether we add x and y together or subtract them.

5.3 Propagation Through Multiplication & Division

Assume I have the following two quantities,

$$\begin{aligned}x &= 24.6 \pm 0.8 \\ y &= 5.4 \pm 0.5\end{aligned}$$

Which one has the bigger uncertainty? Well, obviously $0.8 > 0.5$, so it is tempting to say that there is more uncertainty in x . However, if we think about it more carefully, we have nearly 10% error in y while only having about 3% error in x . So in this sense we have measured x much better than y . It is this *percentage error* that will matter for propagation when we do multiplication and division. Here the result is as follows,

$$\begin{aligned}f_3 &= xy \text{ or } f_3 = \frac{x}{y} \\ \left(\frac{\sigma_{f_3}}{f_3}\right)^2 &= \left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2\end{aligned}\tag{5.4}$$

Note that we are still adding the errors together, whether we are multiplying or dividing. Also we are still treating the errors like vectors and summing the squares of the relevant quantities, but now we are interested in the percentage errors not the errors themselves.

One reason we can justify this is through unit analysis. Remember that it is only legal to add quantities of the same units. So by looking at the fractional error, I am adding together numbers which are unit-less, instead of trying to combine errors of different units.

5.4 Propagation Through Power Functions

Many functions in physics involve raising a measured quantity to a specific power so of course we need a rule for functions of the form,

$$f_4 = x^N$$

where x is the variable and N is a constant that we are raising x to. Now at first glance, it may seem that we don't need a separate rule because we could just use the product rule from above with,

$$f = x^2 = x \cdot x$$

which would give us,

$$\begin{aligned} \left(\frac{\sigma_f}{f}\right)^2 &= \left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_x}{x}\right)^2 \\ &= 2\left(\frac{\sigma_x}{x}\right)^2 \end{aligned} \tag{5.5}$$

However, this turns out to be too small and the reason is because of how we should be thinking about the independence of the errors. In Eq. 5.5 it is assumed that the two errors that are being combined are independent. However, this is not the case when we make a single measurement and then raise that measurement to a power. As an example, let's say you were handed an object, told it was a square, were asked to give the area of the object with uncertainty. Equation 5.5 represents how you should calculate the error if you take the time to measure two sides of the object and multiply them together. By making two separate measurements, you have two independent errors. However, if you were to trust that the object is square and simply measure one side and square it, then the added assumption of squareness adds some uncertainty to that which comes just from the measurement. In this case the uncertainty would be,

$$\left(\frac{\sigma_f}{f}\right)^2 = 4\left(\frac{\sigma_x}{x}\right)^2 \tag{5.6}$$

In general we have the following,

$$f_4 = x^N \quad \left(\frac{\sigma_{f_4}}{f_4}\right)^2 = \left(N\frac{\sigma_x}{x}\right)^2 \tag{5.7}$$

5.5 Propagation Example

On the review assignment quiz you had a question that asked you to estimate the height of a platform based off of the time it took for an object to fall to the ground from that height. On that question it was appropriate to calculate an estimate of the distance from each of the time values given using the equation $\Delta x = 1/2at^2$ and then calculate the statistics on the set of distance estimates. Instead, let's imagine that we know the following,

$$\begin{aligned} g &= 9.81 \pm 0.07 \text{ m/s}^2 \\ t &= 3.25 \pm 0.05 \text{ s} \\ \implies \Delta x &= 51.809 \text{ m} \end{aligned} \tag{5.8}$$

First, we need to get the error in t^2 through use of the power rule, and not just by treating it as multiplication by itself,

$$\begin{aligned} \left(\frac{\sigma_{t^2}}{t^2} \right)^2 &= \left(2 \frac{\sigma_t}{t} \right)^2 = \left(2 \frac{0.05}{3.25} \right)^2 \\ \left(\frac{\sigma_{t^2}}{10.5625} \right)^2 &= 9.467 \times 10^{-4} \\ \left(\frac{\sigma_{t^2}}{10.5625} \right) &= 0.03077 \\ \sigma_{t^2} &= 0.325 \text{ s}^2 \end{aligned}$$

Now we want to combine the error in t^2 with the error in g ,

$$\begin{aligned} \left(\frac{\sigma_{gt^2}}{gt^2} \right)^2 &= \left(\frac{\sigma_g}{g} \right)^2 + \left(\frac{\sigma_{t^2}}{t^2} \right)^2 \\ \left(\frac{\sigma_{gt^2}}{103.6} \right)^2 &= \left(\frac{0.07}{9.81} \right)^2 + \left(\frac{0.325}{10.5625} \right)^2 \\ \left(\frac{\sigma_{gt^2}}{103.6} \right)^2 &= 5.0916 \times 10^{-5} + 9.467 \times 10^{-4} \\ &= 9.977 \times 10^{-4} \\ \left(\frac{\sigma_{gt^2}}{103.6} \right) &= 0.03159 \\ \sigma_{gt^2} &= 3.273 \text{ m} \end{aligned}$$

Lastly, we have to remember that the factor of $1/2$ in our physics equation, so the uncertainty also gets multiplied in half,

$$\sigma_{\Delta x} = \frac{1}{2}\sigma_{gt^2} = 1.636$$

Remembering how uncertainties are to be formatted, we finally have our estimation of the distance with uncertainty,

$$\Delta x = 52 \pm 2 \text{ m}$$

The amount of confidence we have in this range is equal to the amount of confidence we had in the original uncertainties. If both of the values in Eq. 5.8 were given as a 95% confidence interval, then our statement about the distance would also be a 95% confidence interval.

5.6 Summary & Final Comments

Now even with this simple example, there were many steps that were necessary in order to propagate the error. In practice these steps are not used often to do numerical error calculations, as it is generally easier to simply collect a decently large set of results and calculate the statistics on those results. Where this technique is useful is in experimental design, as it allows you gauge which inputs contribute meaningfully to the error in your results. In that way you can plan to use your limited time and money in improving the parts of your experiment that will give you the best improvement.

The propagation rules are re-printed here for quick reference, and I have included the general rule that will work for all functions (*though it is not expected that you will need to use the generalized rule in this course*).

$$f(x) = Ax \implies \sigma_f = A\sigma_x$$

$$f(x, y) = x \pm y \implies \sigma_f^2 = \sigma_x^2 + \sigma_y^2$$

$$f(x, y) = x \times y \text{ or } \frac{x}{y} \implies \left(\frac{\sigma_f}{f}\right)^2 = \left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2$$

$$f(x, y) = x^N y^M \implies \left(\frac{\sigma_f}{f}\right)^2 = \left(N\frac{\sigma_x}{x}\right)^2 + \left(M\frac{\sigma_y}{y}\right)^2$$

$$f(x_1, x_2, x_3, \dots, x_N) \implies \sigma_f^2 = \sum_{i=1}^N \left(\frac{\partial f}{\partial x_i}\right)^2 \sigma_{x_i}^2$$

Lesson 6

Regression Revision

In this lesson we are going to take the intuition that we learned about in Lesson 4 and apply that to our regression analysis. In that lesson we stated that if the value of uncertainty in a result is greater than the value of the result itself, then the result has no statistical significance. This can often happen when running a regression and when it does, it allows us to see what variables are important and which ones are not.

Let's again revisit the sample M2 data for a projectile. The output for the x vs t regression is

3.055189159	-0.000866166
0.010337104	0.004227133
0.999771097	0.010253495
87353.11759	20
9.183796771	0.002102683

Table 6.1: Regression output for a linear fit containing a slope and y-intercept value.

This analysis tells us that the initial x -position was -0.0009 meters, but the amount of error in that result is 0.004 meters. So without even adjusting for a higher confidence interval or for the degrees of freedom, the error is already an order of magnitude higher than the result itself. Remember that this means we have failed to prove that the x -position has a statistically significant, non-zero value.

Now this finding has implications beyond the fact that we would state that the initial x -position is zero (we haven't proven that it's zero, but that's our best guess for it). The regression output would be different if the intercept is assumed to be zero. So we want to run the regression again, but this time we will tell Excel to force the intercept to be zero. Remember that in the *linest()* formula, the third argument has been a "1". This is what has been telling Excel that we would like a regression with a constant term. If instead we type,

$$=linest(y-values, x-values, 0, 1)$$

we will get a regression where the intercept is zero. For the sample M2 data, that would give the following output,

3.053385684	0
0.005222394	#N/A
0.999938572	0.010016776
341840.9726	21
34.29888495	0.002107052

Table 6.2: Regression output for a slope only linear fit.

Note how the value for the x -velocity has changed, albeit slightly. Also notice that because we are no longer estimating an intercept, the number of degrees of freedom has increased by one. Finally, note that the R^2 value is higher in this model than the previous. This is another indication that a model without an intercept value fits the observed data better than a model with the intercept (though this does not always happen).

Now of course the intercept is not the only output in the regression that could have 100% uncertainty or more. It is certainly possible that one or more of your potential x -variables is not needed either. In this case you would simply run the formula again, but only highlight the x -variables that are significant. You would repeat this process until only significant variables remained. As long as the intercept remains significant, you would keep the third argument in the *linest()* formula as "1".